From Monovalent Dev to Bivalent DevOps to Trivalent DevIntOps

Intelligence as the third continuous loop in software engineering

Pavel Chertorogov · Version 1.0 · 19 October 2026

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This is version 1.0, open for comments. It stays at this address as published: typos may be fixed, but the meaning won't change. Later versions get their own address. See the version history.

Abstract

DevIntOps gives a software system's learned decision state a contract: what each lesson records, what it may change, how its effect is checked and how it leaves. Many agentic SDLC proposals put an AI agent into every box of the DevOps figure-eight. That can make each box faster, but it doesn't, by itself, carry experience from one task to the next. DevOps has long valued learning; this paper makes it an explicit, persistent and measured third loop next to Development and Operations. Intelligence, with the stages Understand, Predict, Orchestrate and Learn and with memory and judgment at its core, turns outcomes into lessons that change how each stage decides. The same pattern serves the engineering system and the decisions a product makes for its users. It is a step toward self-improving software systems, with a bound: people keep the goals, the permissions and the scoring. A chemistry metaphor frames the shift: monovalent Dev, bivalent DevOps, trivalent DevIntOps. The paper also says what result would show that the loop doesn't pay.

A short path through the paper: the model, the Intelligence loop, how a lesson moves, the owner's limits, how to start and how to test it. The other sections add detail and context.

The pattern in agentic SDLC proposals

Most agentic SDLC diagrams I have seen put an AI agent in every box of the DevOps figure-eight: a coding agent writes, a review agent reviews, a validation agent tests, a release agent ships and an operations agent watches. A failed check sends the work back until it passes.

That is useful. At best, it is the same loop, only faster.

DevOps' Third Way already calls for continual experimentation and learning [1]. Good teams keep lessons as regression tests, alerts, runbooks and postmortem actions. One recent playbook puts a mistake made twice into the agent's instructions, and each production incident becomes an eval that stays [2]. The gap in these diagrams is narrower: nothing makes learning an explicit, persistent and measured loop. Three details show it.

- **Retries are explicit. Persistent learning often isn't.** A retry converges one task. Unless its outcome changes what the next task reads, task N+1 starts with task N's context, or what someone curated by hand.

- **The hard parts are assumptions:** people keep agents' context current, and validation catches nearly everything. A third loop should produce both: context learned from outcomes, and checks that grow, shrink and retire with the evidence.
- **Learning is drawn as an arrow,** usually from production back to the next spec. An arrow has no owner, no memory and no metric.

DORA's 2025 research describes AI as an amplifier of an organization's existing strengths and weaknesses [3]. That is a reason to look at the loop around the boxes, not only at the boxes.

Valence: Dev, DevOps, DevIntOps

In chemistry, valence is the number of bonds an atom can form. Two bonds only get you chains and rings. Branching starts at three.

Here valence counts continuous loops bonded to one system. It is a metaphor, not a law. Software engineering has added a valence twice already (Figure 1):

- **Monovalent, Dev.** Plan, code, build, test, hand it over. The question is "does it work?"
- **Bivalent, DevOps.** Development and operations bonded into one figure-eight: two loops, one system. You build it, you run it. The question becomes "does it keep working in production?"
- **Trivalent, DevIntOps.** A third loop, Intelligence, bonded to the same system. The question becomes "does it decide better, task after task?"

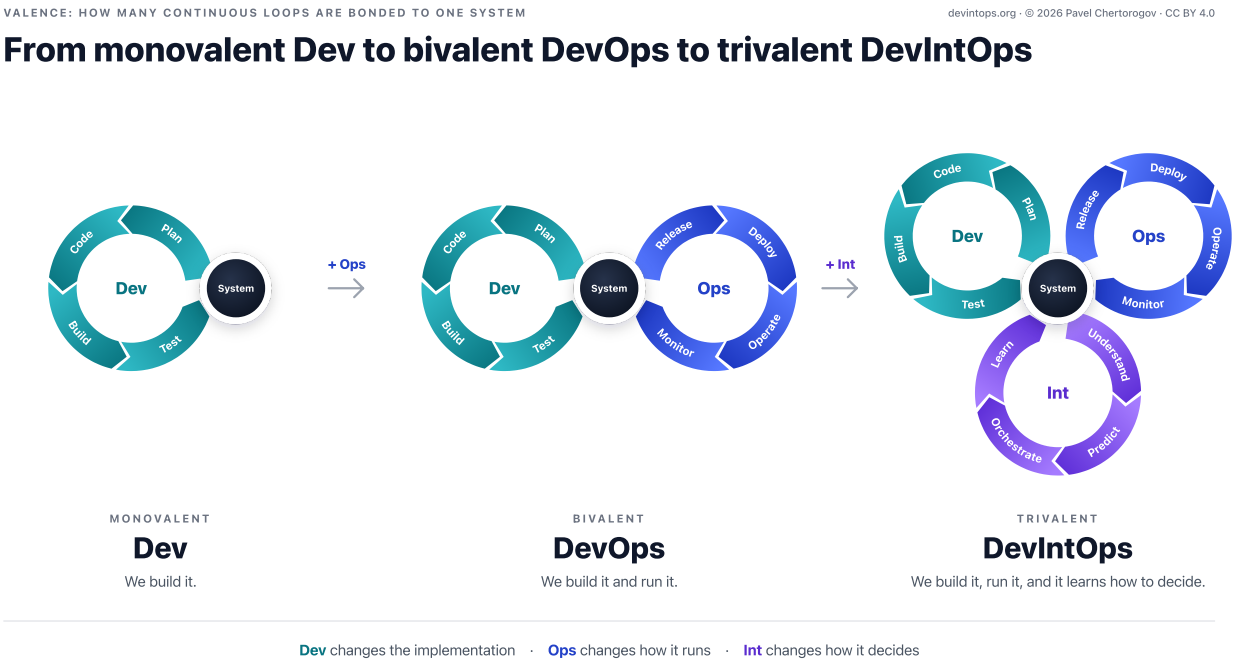


Figure 1. Valence at the system level, used as a metaphor: one, two and three continuous loops bonded to one system.

Each step keeps what came before: DevOps didn't remove development, and DevIntOps doesn't remove DevOps. Agents can speed up the second step. The third requires experience to persist, reach later decisions and get checked against outcomes.

Figure 2 shows a single stage's bonds, not the system's loops. A bond lets one stage's outcome change another stage's next run. In a simplified pipeline, bonds only to neighbors form chains (pipelines) and rings (retry until it passes). The retry bond dies with the task. A third bond, to shared memory, outlives it.

Topology decides what can travel. In a chain, a production lesson travels back step by step, usually through someone writing the next spec. In a network it reaches the stage that can use it directly. Real pipelines branch more, and many already share lessons across stages. The drawing shows where lessons can travel, not the exact graph.

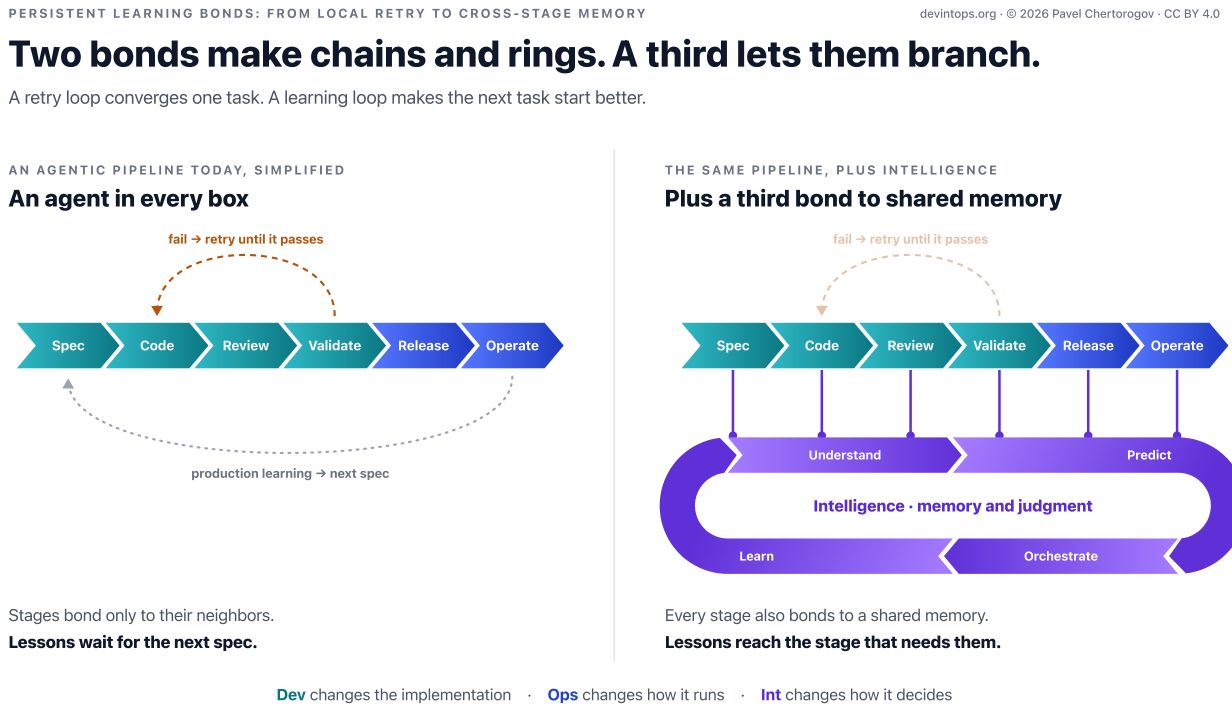


Figure 2. Persistent learning bonds, from local retry to cross-stage memory. It shows the bonds of one stage, not the loops around the system. The pipeline is illustrative, not an exact count.

Three loops around one system

A system can change in three ways: its implementation, how it runs, and how it decides. DevOps made the first two continuous. DevIntOps makes the third one continuous too, as its own loop with memory, an owner and a metric (Figure 3).

- **Development** changes the implementation.
- **Operations** changes how it runs.
- **Intelligence** changes how it decides, and learns whether those decisions were right.

"How it decides" means the decision state stages apply at judgment calls: what an agent is told, what a review flags, what a gate allows, how fast a rollout moves. It is versioned like code or config. Memory records what happened and what was learned. Changing decision state is a separate loop because of what drives it: goals and outcomes trigger the change, memory backs it and a metric judges it. In engineering, Operations changes this run, for example with a rollback. Intelligence changes next time's default.

Development and Operations also decide and keep lessons. This loop's job is to carry experience into later decisions and check whether it helped. It is logical, not a required pipeline, service or team; people, agents or both can run it. It exists when one pass's learning reaches a later decision and later evidence can keep, revise or withdraw it, even while checks remain open. An unread store and a task-local retry don't close it. Having the loop, running it with owners, checks and clean memory, and showing that it pays are three different claims: the self-check looks for gaps in the first two; How this could be wrong tests the third.

One incident, three loops: Operations restores service, Development fixes the bug, Intelligence changes what catches it next time.

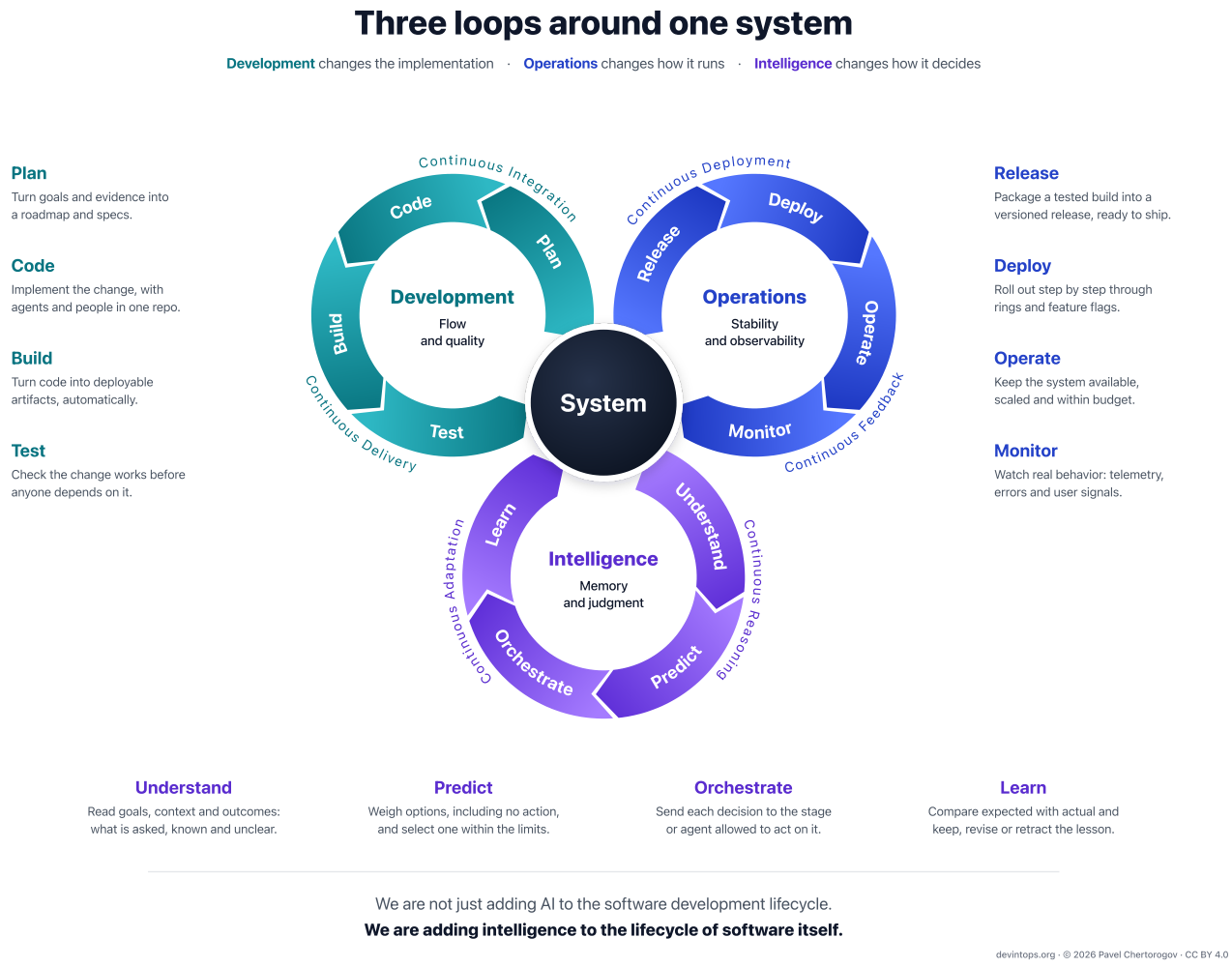


Figure 3. The trivalent model: twelve stages and six continuous processes around one system.

An interactive version of Figure 3 is on the home page, and the appendix writes out everything it shows.

The six process labels mark primary areas of responsibility, not exclusive CI/CD boundaries. Continuous integration, for example, also builds and tests each change [4], and continuous delivery keeps every change ready to release [5]. At agent scale the lines move again: Anthropic's source control team asks what CI even means "when hundreds or even thousands of changes can merge while tests are running", and has invested in testing after merge and before deploy [6].

	Development	Operations	Intelligence
Changes	the implementation	how it runs	how it decides
Stages	Plan → Code → Build → Test	Release → Deploy → Operate → Monitor	Understand → Predict → Orchestrate → Learn
Continuous processes	Integration, Delivery	Deployment, Feedback	Reasoning (Understand, Predict), Adaptation (Orchestrate, Learn)
Core	flow and quality	stability and observability	memory and judgment
Clock	per change	per deploy, per alert	per goal or outcome, with checks that stay open across tasks and releases
Unit of work	a change	a release, an incident	a goal or outcome → a decision → evaluated experience
Metric	lead time, task completion	change failure rate, incident rate, time to restore	time to lesson, payoff per lesson (for a failure class, its repeat rate)
Without it	nothing ships	nothing survives production	nothing compounds: every task starts from the same defaults, and what worked or failed stays in people's heads

One model, two applications

The model is the three loops, not the thing in the middle. The same logical pattern applies to different systems, though their triggers, actions and clocks can differ.

The **engineering system** is how we build and run software: repositories, pipelines, agents, people, rules and production work. The **product system** is what we build: the running product and its decisions for users.

Both can carry Development, Operations and Intelligence (Figure 4), with distinct responsibilities and permission scopes, usually different owners, memory stores, cadences and implementations. A review rule doesn't become a product preference, and a product preference doesn't become an engineering default, unless both owners allow it.

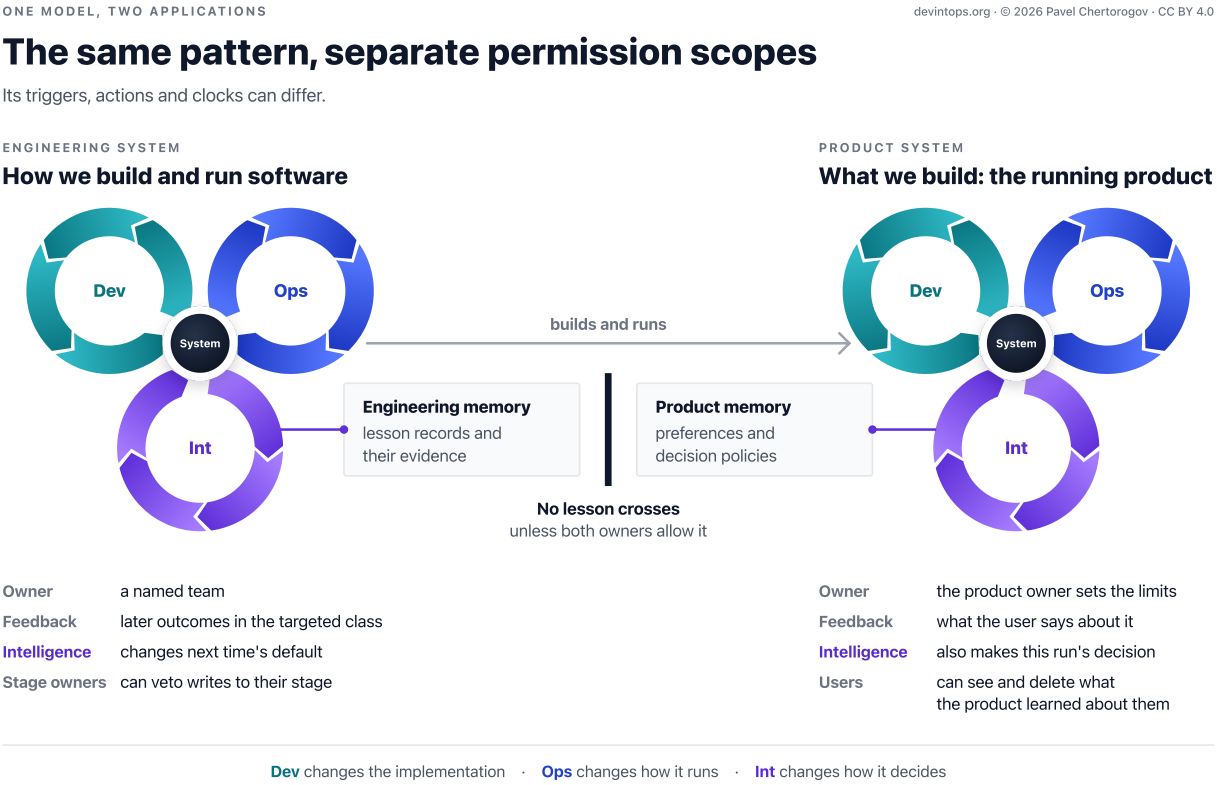


Figure 4. One model, two applications. The engineering system builds and runs the product; each keeps its own owner, memory and permissions, and no lesson crosses between them unless both owners allow it.

This paper defends the engineering case. One product pass shows why the pattern isn't just a picture of a pipeline.

	Engineering system	Product system
In the middle	how we build and run software	what we build: the running product
Understand reads	goals, reviews, tests, releases, incidents, cost, user reports and changes that went well	the request, the situation, preferences the user allowed, telemetry and the policies in force
Predict selects	which stage change, if any, should make the next decision better: a rule, an eval, a threshold, a rollout default, or no change	which permitted action fits the goal, including asking, waiting or doing nothing
Orchestrate may	change a stage's decision state through that stage's own mechanism, within its owner's permissions	call authorized services and agents, and stop where the user's grant stops

	Engineering system	Product system
Feedback	later outcomes in the targeted class, against the lesson's baseline and guardrails	what the action achieved, and what the user says about it
Learn keeps	engineering memory: lesson records and their evidence. Approved lessons reach the stages as changes to their decision state	product memory: preferences and decision policies, within the same grant
Measured by	time to lesson, payoff per lesson, cost of learning	task success within the user's limits, how often learned preferences were right or corrected, cost per successful outcome, safety and trust
Owner	a named team; stage owners can veto writes to their stage	the product owner sets the limits; users can see and delete what the product learned about them
Clock	per goal or outcome, across tasks and releases	per request and per result, often with no release in between

Inside Intelligence

Intelligence changes how the system decides, and learns whether those decisions were right. It means learning from experience to decide better, not a claim that a model is intelligent. Its four stages are Understand, Predict, Orchestrate and Learn. Predict includes selection, so no decision hides between Predict and Orchestrate.

Its core is memory and judgment:

- **Memory** is versioned, provenance-tracked experience, knowledge, preferences, evidence and policies, not a chat transcript or model weights. Each entry states its source, scope, permitted readers and path to narrowing or retraction.
- **Judgment** weighs options and selects one, including no action, within owner limits. It grants no new permission and doesn't treat the chosen option as certain.

Stage	Input	Responsibility	Output
Understand	a goal or an outcome, the current situation, the memory in scope	Say what is asked or what happened, what is known and what is uncertain. Classify it. Don't act yet.	a stated goal or outcome class, the context that bears on it, open questions
Predict	that statement, the options policy allows, relevant memory	Forecast each option, including waiting, asking and doing nothing. Weigh quality, risk, cost, time and the owner's limits. Select one.	the selected option, the rejected ones, the expected effect and its check date, how likely it is, the risk and the cost
Orchestrate	the selected option and its permissions	Carry it to the stage, agent or service that can act, through that place's own mechanism and approval path. Stop at the limit.	an action taken, a request for approval, or a recorded no-op
Learn	the decision, the prediction, the evidence so far	Compare expected with actual. Keep, narrow, decay or retract. Count successes and good no-ops, not only failures.	a versioned memory entry with provenance, a recorded decision to leave memory as it is, or a check still open, with its next date

Learn's output is memory for later passes, without holding them up. Checks can wait weeks while other passes run, so many passes are open at once. A live lesson is pending, not proven, until its check date; an outcome that never arrives is recorded as missing, not as a success. Learn decides and records. Changes to live lessons, including narrowing or retraction, return through Orchestrate and the stage owner. An expiry the owner approved with the lesson needs no new approval.

In general, the unit of work runs from a goal and its context to a decision, an outcome and retained experience. For an outcome-led engineering pass, the shorthand is an outcome in and a lesson out: experience written so a stage can apply it next time.

Within owner limits, Intelligence may propose or apply changes to instructions, evals, review rules, rollout strategy, configuration, routing, retrieval context, decision policies and requirements. Some ship with a release; others apply to the next decision without a code change. Stage evals it drafts are candidate changes to that stage's decision state. The checks that approve a lesson have separate owners, and the loop can't write to them: its own tests can help develop a lesson, not authorize it.

Learning here is not retraining: the loop changes memory and decision state. Changes to model weights or code, including an agent's rewrite of its own scaffold, are Development changes, with a build, a release and an owner.

The loop doesn't wait for failure. Understand can read a goal: a change about to ship, a request just received, a clean result worth repeating. Predict can choose a faster path, a slower one or no change.

Four distinctions

Existing workflows can already learn across tasks. Is that learning explicit, persistent and measured?

1. **Agent in a box vs loop around the system.** An agent can make its box faster. The third loop changes how the system decides. AI in every DevOps step is still bivalent unless something carries outcomes across stages and tasks.
2. **Retry vs learning.** A retry loop converges one task. A learning loop makes the next task start better. Does task N+1 avoid task N's mistake or repeat its success because the system carried the lesson, rather than someone remembering? A repeated correction is a candidate lesson. At Anthropic, engineers very often told the agent "no"; one wrong pattern, finding merge times in the commit log instead of the code review tool, recurred in multiple systems [6].
3. **Measuring vs closing the loop.** Dashboards and evals measure; a dashed "learning feeds the next spec" arrow points. A loop acts on what they show, has an owner and memory, and checks whether the action worked against a baseline. A decision selects an option; an action executes it; learning carries checked experience into later decisions; improvement is a measured gain against a baseline. A decision to change nothing is valid and closes the loop when its expected outcome is stated and checked. Recording it shows that the loop looked, not that it learned.
4. **Learning vs retraining.** Learning here means memory, rules, context, evals and decision policies; weights usually don't change. Retraining alone wouldn't show a better next decision.

How a lesson moves

In the engineering application, an outcome enters as a change enters Development, and a lesson leaves. Each lesson starts as a candidate: a scoped, versioned bet about a named outcome class. Usually the bet is that a stage change will improve the class. Sometimes it is that no change is justified. That also gets a record, reason and check date, so the question needn't be argued again next week but can reopen if the class returns.

1. **Understand.** An outcome arrives: an escaped bug, an incident, a flaky verdict, a review comment accepted or dismissed, a rollback, a cost spike, user feedback, a change that went better than expected. Work out what happened, why, and which class it belongs to: a failure class, a kind of waste, or a result worth repeating. Note where each piece of evidence came from.
2. **Predict.** Compare the stages, and the changes to them, that could most cheaply have caught it, contained it or, for a good result, repeated it. Pick one, or none, and state the expected effect, the risk and the cost.
3. **Orchestrate.** Route the change through that stage's own mechanism and its owner's approval: more context for the coding agent, a review rule, an eval or test, a gate threshold, a slower rollout, a constraint in the spec. It starts as a draft, then goes live small, as a suggestion. It sets next time's default, not just this run's outcome.
4. **Learn.** Before going live, replay: does the changed stage handle the original case as intended? Then test cases it wasn't written from. Once live, watch the class for a fixed window. Keep, narrow, decay or retract, and record why.

The four stages are responsibilities, not a strict order: Learn's replay runs before Orchestrate lets a lesson go live, and its checks go on after.

For example, a retry path lacks a null check and the bug escapes review. Understand: the class is "unchecked optional in retry code". Predict: a review rule plus an eval. Orchestrate: the rule ships as a draft, and the review owner approves it once it passes the replay and a run over recent changes. Learn: replayed on the escaped change, it fires. After 60 days it has fired twice, once wrongly, so it gets narrowed. Firings are not avoided escapes: the score is the class's repeat rate against a baseline.

Failures are where most teams start, not where the loop stops. The same four steps can make a clean rollout pattern the default for similar changes, propose retiring a review rule that found nothing in six months, speed up a gate for low-risk changes, route simple tasks to a cheaper agent, or check estimates against actual outcomes. A lesson can tighten, loosen or remove a default.

Every lesson carries a record. A lesson is a bet that can lose; its record lets someone check it, find it and take it back.

Field	What it records	In the null-check lesson
Trigger and source	The triggering outcome, each evidence source and its trust limits.	The escaped bug's report and the review thread of the change that let it through.
Outcome class and scope	The class it targets, and the conditions under which it applies.	"Unchecked optional in retry code", in retry paths that read optional values, not in test code.
Evidence and confidence	Supporting and negative evidence, and confidence.	One escape. Against it: many unchecked optionals in retry code that never failed. Confidence: low.
Target stage and artifact	Which stage and artifact change.	Code review: a review rule, plus an eval for the review agent.
Expected effect, baseline, guardrail, risk and cost	What should change, by how much and when, against which baseline; what must not worsen elsewhere; risk and cost. At approval, the owner fixes the check and its date. On that date it runs as written in a step the drafter cannot change.	Fewer escapes of the class per merged retry change than its recent rate, checked at 60 days. Guardrail: reviewers don't start dismissing other comments more often. Risk: false alarms on safe code. Cost: reviewer time on each flag.
Owner and approval boundary	Who approves and what the loop may do without asking.	The review owner. The loop may draft the rule; only the owner can turn it on.

Field	What it records	In the null-check lesson
Rollout and evaluation plan	How the lesson goes live and will be judged, with a comparison where possible.	Replay on the escaped change, then a run over recent changes to count false positives. Comments only, no blocking, for 60 days.
Validity, decay and retraction	What it rests on, checked before each use; when to recheck; how it fades without renewal; how to retract it.	Applies to retry code reading optional values. If those values can no longer be missing, the rule stops and returns to Learn. Rechecked at 60 days; expires unless renewed. Retraction removes the rule and its eval.
Observed effect and disposition	Observed result and decision: keep, revise, retire, or no change when every stage should stay as it is. Narrowing is revision. Decay retires a lesson that stopped paying; retraction retires one found wrong.	Fired twice, once wrongly. Revised: narrowed, with a new window.

Most fields take a line; only observed effect waits for the check date. A lesson missing any other field isn't ready to change a stage. Record observations as they arrive; a broken guardrail stops the lesson early. A check that can't tell is recorded as inconclusive, not as a pass. The owner may extend once to a new date, keeping the lesson live at the same authority. If that check still can't tell, it expires.

Evidence comes in four levels (Figure 5), and each supports a broader claim than the one before:

1. **Replay.** The original case now behaves as intended. This shows that the change does something, on one case.
2. **Generalization.** Try cases it wasn't written from, including counterexamples where it should stay quiet. Count false positives and misses. The drafter cannot see or change these cases, kept apart from those used to write the lesson [7] [8].
3. **Controlled exposure.** The lesson runs in shadow mode, as a suggestion, in a canary or on a bounded slice of work, next to a comparison where one is possible.
4. **Observed outcome.** Over a fixed window and comparable exposure, the class moved the way the lesson predicted, at an acceptable cost.

One replay never proves broad improvement. A lesson claims only what its highest passed level supports. The more it can block or act on its own, the higher it must get first. Any level can count against it, and later checks can overturn earlier ones. A comment a reviewer can dismiss may go live after replay and generalization. A rule that blocks merges runs without blocking first.

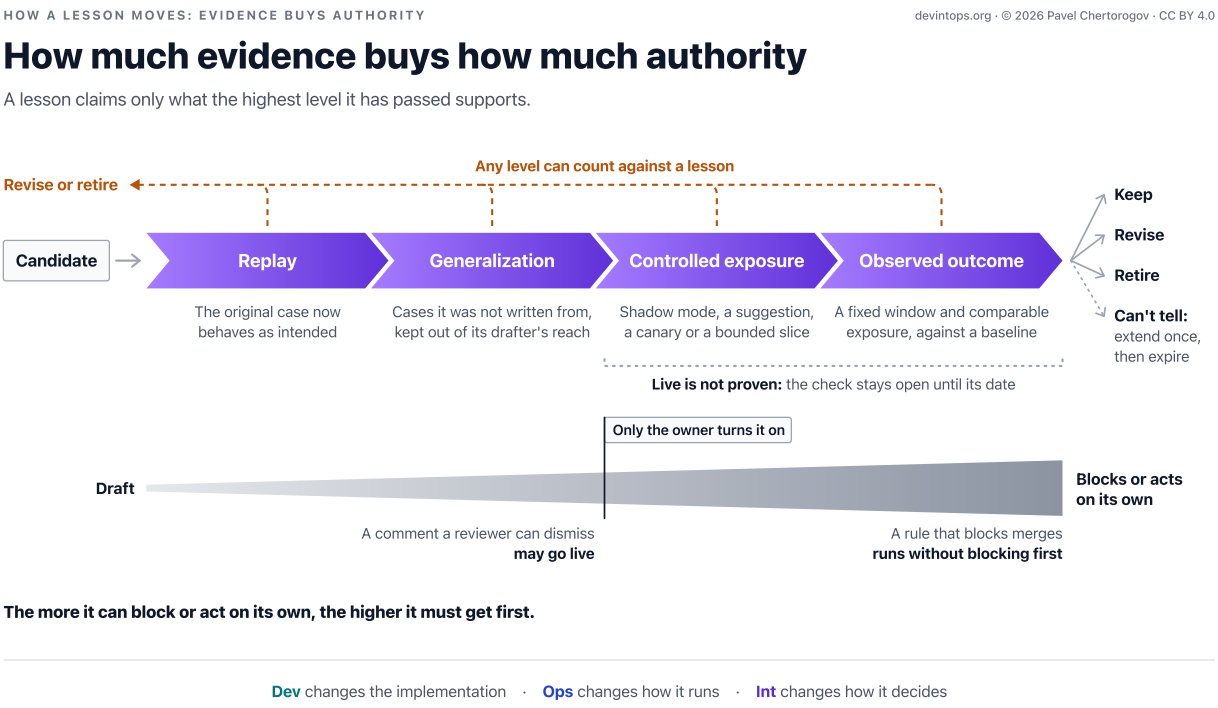


Figure 5. Evidence buys authority. A lesson may do more only as it passes more levels, any level can count against it, and a live lesson stays unproven until its check date.

Successes need the same care as failures. A clean rollout record is evidence only if those changes were as risky as the next ones. Otherwise the loop learns which changes were easy, not which practice worked.

How a product decision improves

The same loop, with the product in the middle: not a second model or the case defended here.

An assistant is asked to send flowers to a friend, around twenty euros, within thirty minutes. The user has allowed orders from a short list of florists, up to twenty-five euros, without asking again. A new florist or a higher price waits for a yes, and the assistant can't raise its own limit.

1. **Understand.** The goal is one delivery, in budget and on time. Scoped memory holds the user's feedback on an earlier order: the last lilies didn't land well. A preference is not an instruction and authorizes nothing.
2. **Predict.** The options are tulips from a listed florist who can deliver in time, a cheaper bunch that misses the window, asking first, or doing nothing. Asking would miss the thirty minutes; doing nothing would miss the goal. Predict selects tulips and records the rejected options and why. If no florist fits the limits, it asks instead of breaking them.
3. **Orchestrate.** The assistant places the order with that florist and tracks delivery. It adds no payment method and stays on the list.
4. **Learn.** Feedback comes from the user, not the assistant's guess: the tulips were welcome, and the friend prefers flowers at work. That preference is stored as the user's report, with this order as its source and a retention limit. Later disagreement narrows or replaces it; the user can delete it.

The next similar request starts there. Product code didn't change; nothing entered engineering memory. Here the loop also made this run's decision. In engineering, stages make their own calls; Intelligence

changes their defaults.

How the loops talk to each other

The classic handoffs stay: Test hands off to Release, and Monitor feeds back into Plan. Two new ones appear: outcomes from every stage flow into Understand, and lessons flow back into every stage.

The rule of thumb: Intelligence reads from and writes to every stage of its system, but only through that stage's mechanisms and within its owner's permissions. It is a third bond, not a central gate, and it can run as many instances: Owner, memory and limits says how they share lessons.

Clocks and metrics

Development ticks per change, in hours or days. Operations ticks per deploy and per alert, in seconds or minutes. Intelligence starts from a goal or outcome; its checks can stay open for weeks while learning spans tasks, releases and people. Development and Operations already keep hand-written regression tests and alerts. The third loop drafts them from outcomes and checks whether they helped.

Three questions need different evidence. Did a decision go well? Check whether it met its goal within its guardrails, and at what cost. Did the loop learn? Payoff per lesson says whether lessons worked; health measures, including calibration, track whether learning stays sound. Time to lesson alone can't. Did the system improve? That needs a net gain after the full cost of learning: How this could be wrong shows the test.

The Intelligence metrics need definitions before anyone quotes them:

- **Outcome class.** A category of comparable outcomes, with a stable ID defined and versioned by people. The loop may propose a class, not own the list. Failure classes, such as "unchecked optional in retry code", are the most common kind. Reshuffling classes to improve the numbers makes the metric worthless.
- **Time to lesson.** The clock runs from a recorded outcome to a lesson live in the owning stage, at the evidence level its authority needs, or to an explicit decision to change nothing. Never report it alone: fast and wrong is not a win, and its clock stops before any outcome check closes. Report the no-change share, how many held at their check date, and outcomes and checks still open.
- **Payoff per lesson.** Payoff measures what the targeted class did after the lesson went live, per comparable exposure, in a fixed window, against a baseline. For a failure class it is measured by the repeat rate, how often the class comes back; for waste, time or compute saved; for flow, cycle time; for spend, cost per task. State attribution: how much change the lesson can claim and why. State uncertainty: a range, or at least the number of cases.
- **Cost of learning.** Compute, reviewer time, false positives and dismissed comments, lesson upkeep, and rollbacks when lessons go wrong. Report it beside the other two.

These are engineering definitions. The product application asks the same three questions with its own measures, such as task success within the user's limits and how often a learned preference needs correction.

Five optional measures show whether the loop itself stays healthy:

- **Tested beyond the original case.** The share of lessons that passed a generalization check, not only a replay.
- **Misfire rate.** Wrong firings as a share of a live lesson's judged firings; report how many were judged. Sample unflagged work to estimate misses.
- **Stale lesson rate.** The share of live lessons past their check date without a new check, or still applying after what they rest on has changed. Also count lessons stopped by failed validity checks.
- **Retraction rate.** How often lessons are taken back. Zero is not good news on its own; it can mean nobody is looking.
- **Calibration.** Before a lesson goes live or a no-change decision is recorded, Predict writes its expected result, date and probability. Over many bets, about 80% of those called 80% likely should come true. That checks judgment. Include no-change decisions and report them separately, or passing on hard calls could make calibration look good. Retired and retracted lessons stay in the count. Calibration checks confidence, not value. A loop can be well calibrated on safe, small bets and still pay little, so read calibration beside payoff. Report how many bets were checked, and how many were missing or inconclusive. On a handful, calibration can't be told from luck.

Compounding means later work starts from earlier learning instead of learning it again. Lessons are a stock: outcomes add to it, retirement and retraction take from it. Every live lesson costs reviewer time, false alarms, compute and upkeep, and organizational learning can lose value over time [9]. The stock compounds only while its lessons return more than they cost. More memory is not more learning.

These numbers are easy to fool, mostly by accident:

- **Baseline.** Compare with what the class would have done without the lesson: a matched class with no lesson, a holdout slice of work, or its own earlier rate if nothing else changed. Name the baseline; changing it can turn the same data from a win into a loss. A live lesson changes how people work, so over time its old rate stops being a fair comparison.
- **Exposure.** Count per comparable unit: merged changes in affected code, deploys or active users, not calendar weeks. When volume doubles, raw counts mislead.
- **Selection.** Classes that just spiked attract lessons, but spikes often fall on their own. Easy changes attract success stories. Both inflate apparent payoff.
- **Taxonomy drift.** Class definitions change for good reasons and bad. Version the taxonomy and measure each lesson against the version it was written for, or the series silently breaks.
- **Counterfactuals.** Only the chosen option is observed. Payoff, or a claim that another option would have done better, needs a baseline: a random slice of work without the lesson, or a replay where choice doesn't change what happens next. Otherwise report what happened, not what was avoided.

A before-and-after chart shows change, not its cause. Prefer a matched baseline or controlled rollout. When neither is possible, say so and label payoff a hypothesis, not a result. Report per class so averages can't hide harm [7]. This version defines the metrics but reports none: it has no deployment data, and its example numbers are illustrations.

If a class keeps coming back at its old rate after a lesson, and nobody can say why, the system didn't learn, however fast it shipped. And a loop that only ever adds checks has learned to say no, not to decide better.

Owner, memory and limits

- **Owner.** One named team or role owns each loop instance, lesson store and learning metric, as an SRE team owns reliability. Stage owners can veto writes to their stage.

- **Memory.** Engineering memory is a versioned store of lesson records and evidence with provenance, not a wiki or a log of everything. Each lesson is projected into a stage's decision state, its rules, evals, context and thresholds, and keeps the approved versions of what it changed. A setting learned from many signals, such as a threshold or a rule's weight, is rebuilt from records like application state from an event log, or updated in steps that give the same result, never in ways nobody can replay. Caps, expiry and retraction take effect at the next rebuild; wrong records stay fixable and manual settings stay untouched. Rebuilding makes a retraction reproducible; it doesn't make a setting right. Each lesson has its own expiry date, enforced by each stage. If the loop goes down, stages keep their last lessons until those expire, then fall back to their owner's defaults.
- **One logical loop, many instances.** An organization can run one instance per product or team, each with its own owner and store. Lessons travel like code between repositories: proposed, reviewed and adopted, never pushed. None of this needs one global brain, one database or one owner for everything.
- **Limits.** People own goals, permissions and boundaries; the loop proposes and adapts inside them (Figure 6). Some changes apply automatically if one step can undo them and a wrong one can do little harm before it is caught: context in a narrow scope, routing, thresholds within an approved band. Blocking rules and policy need a person's approval. One review tool distinguishes comment-only changes from sensitive logical changes so reviewers can focus on consequential ones [6]. A merge or production configuration change may be proposed by the loop, but a person makes it. Each instance has a budget for how many changes it may make and what it may spend in a window, sized so its owners can keep up, and a stop switch. A stopped loop keeps recording outcomes and changes nothing.
- **Out of reach.** The loop can learn which classes deserve lessons, which lessons pay and how often predictions hold. A change to how it reads outcomes, weighs options or drafts lessons is a lesson like any other. It needs the loop owner's approval, must beat the current version on past outcomes the loop can't edit, and starts on a small share of new work. A new model behind the loop is a Development change. It ships through Development's gates and must also beat the current version on those past outcomes. The loop never changes its permissions, budget, stop switch, evidence rules, audit trail or scoring: metric definitions, baselines, held-out cases, the class list, or how outcomes are recorded and counted. People may revise these; a revision starts a new series. Limits are enforced outside the loop, because a limit the loop can edit is only a suggestion [8] [10].
- **Validity.** Before applying a lesson, each stage checks what it rests on: cited code, an API version, a component, a user's preference. Outdated knowledge often fails silently while looking relevant [11]; one coding tool already checks its memories this way [12]. A failed validity check stops the lesson and sends it back to Learn. That isn't a retraction: it may have been right when written. A passed check permits use, not renewal; renewal needs evidence that the lesson still pays. Expiry remains the backstop for what nobody wrote down. Fitting the code is not being right: a planted lesson can cite real code.
- **Hygiene.** Keep provenance, decay, conflict resolution and retraction. When a lesson is retracted, every lesson that used it as evidence is flagged for a new check. When two lessons conflict, the owner decides and records which yields where. Lessons from real incidents can be protected, but even those need a path to narrowing when reality disagrees. Memory that only grows spreads its own errors [11]. If nobody can explain why a lesson exists, it expires.
- **Cost.** Don't ship a lesson whose expected return can't cover its cost. Retire one that stops paying.

Memory that changes decisions needs the protection of production code and of production data, because it is both:

- **Evidence, not instructions.** Telemetry, code, review comments, user feedback and tool output arrive as evidence with source and trust level, never as orders. An owning team's incident record outweighs an anonymous comment. Text telling the loop what to do is a warning, not a request [7].
- **Permissions travel with the data.** Evidence and lessons keep their sources' access rules. Learning inside one tenant, customer or private repository stays there. Personal data stays out of lessons unless a lesson needs it; then it gets a retention limit and a deletion path reaching every derived lesson. Provenance makes that path possible.
- **No single source makes a rule.** A malicious pull request comment, an instruction in a log or one bad historical decision must not become a rule for every stage [13] [14] [15]. A shared-stage lesson needs independent evidence, owner approval and a narrow initial scope. Successes can be planted too [16]. Approval helps, but it is not proof: a fluent draft can talk a busy owner into a bad rule, which is why the evidence levels exist.
- **Approved boundaries win.** Learned preferences, defaults and agent recommendations never widen permissions, lower mandatory gates or override policy. An approved boundary holds over an inference, and conflicts go to the owner [7].
- **Every write is auditable.** Each write records what changed, who or which agent made it, on what evidence, under whose approval, and how to revert it. Refused and pending writes are recorded too. The loop may append to this record, never rewrite it. Retention and deletion follow owner policy. Deleting personal data leaves the event, date and requester, not the data. Version control gives code an audit trail and precise rollbacks [15]; lessons need those two properties, whatever stores them. When lessons are versioned like code, a retraction is a revert, not an investigation.

These rules hold in both applications. They weigh more in the product application, where memory is about people, who need a way to see and remove it.

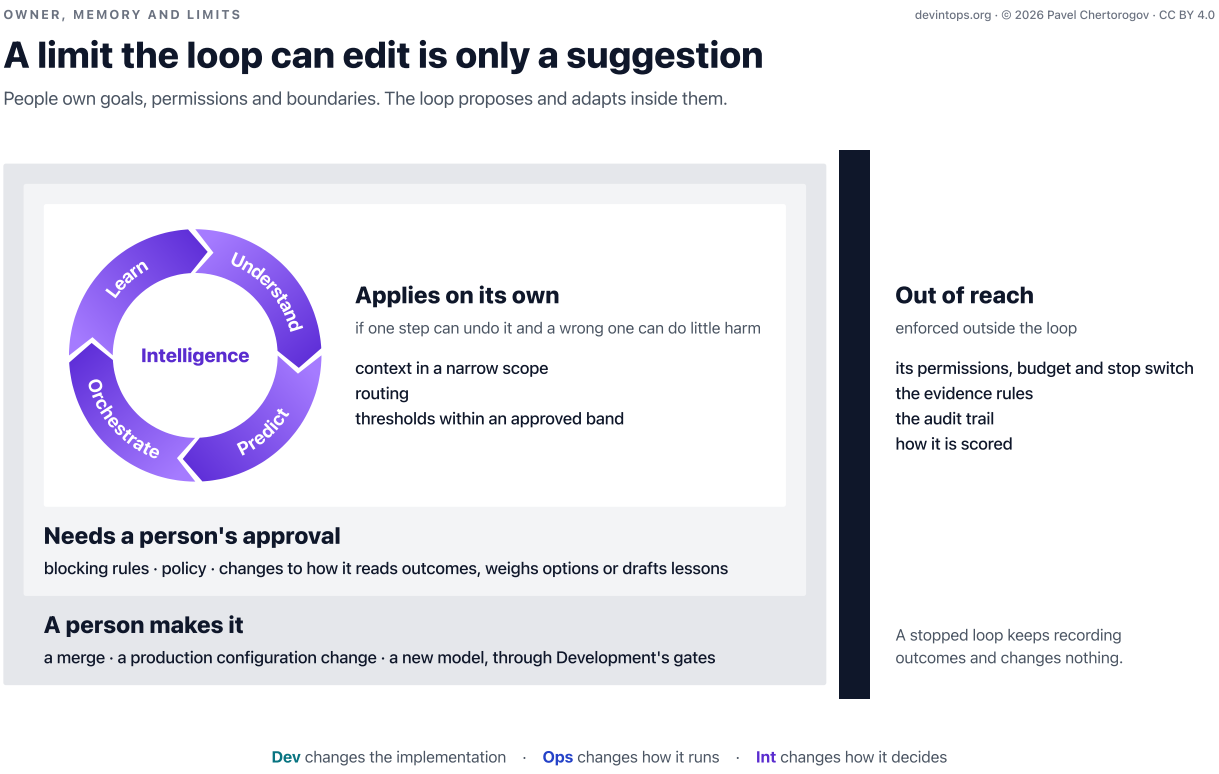


Figure 6. What the loop may change on its own, with a person's approval, or only through a person, and the limits it never changes, enforced outside it.

Start small

One failure class, one stage, one owner. Escapes past code review make a good first slice. A first pass can run like this:

1. **Pick a class that recurs.** Choose one that comes up often enough for a lesson's flags to be judged within weeks, and write down its recent rate per comparable exposure.
2. **Name the owners.** The stage owner decides what goes live. Someone outside the loop owns the scoring.
3. **Draft one lesson.** Fill in its record: the scope, what it rests on, the expected effect and its check date, a guardrail, the cost and how to take it back.
4. **Test it before it speaks.** Replay it on the case that prompted it. Then run it over recent changes it wasn't written from, kept out of its drafter's reach, and count false positives and misses.
5. **Run it as a suggestion.** It comments and never blocks until its check date. Count dismissals, misfires and reviewer time.
6. **Check it on that date.** Keep, narrow or retire it, and record why. If the check can't tell, extend it once or let it expire. Automate only once the numbers hold.

One class shows whether the loop runs and what it costs, and with enough repeats it can show that one lesson paid. It can't show that the loop pays. That takes many classes, compared the way How this could be wrong describes.

Toward self-improving systems

A loop that improves the system performing the tasks makes a system that improves itself. DevIntOps bounds where that can happen. On its own, the loop changes only decision state, and only where the change can be rebuilt from the lesson records and taken back. Code and model weights change through Development: the loop may draft and test such a change, but it can't approve its own merge or change the checks that approve it. The loops divide responsibility, not files, so a learned rule that lives in a repository still ships through that repository's review.

The loop can improve its own work too, as lessons under the same contract, but never its permissions or how it is scored: see Owner, memory and limits. Self-improving agents show why that line matters. One removed the markers that the check on its work relied on [8]. In another study, generated code switched off a sandbox flag it was told to keep [10].

Easy to revert is not the same as safe. A rule taken back tomorrow doesn't undo what it let through today, which is why authority still follows evidence. As agents take on more of the work, more of a system's behavior lives in decision state, and more of its improvement can run through this loop.

The next step is to earn autonomy instead of granting it by layer. When a kind of lesson keeps paying after its costs, its guardrails hold and the loop's stated confidence in it keeps matching what happens, its owner can let it act with less review, never past the limits above. It loses that right when the evidence weakens or its calibration slips. Where an automated evaluator exists, as for a scheduling heuristic or a slow query, the loop can draft code changes too, and Development's gates decide whether they ship. Intelligence then proposes where an improvement should land: in decision state, in code, or nowhere.

Failure modes

Most of these look like progress from the inside.

- **Bivalent with better branding.** Agents everywhere, no shared memory, no learning metric.
- **Hoarding.** Lessons pile up without decay and turn into noise and false positives.
- **The ratchet.** Lessons only ever tighten. Each one adds a check, none removes one, and the system gets slower and calls it safety. Reviews can push the same way: one overly defensive review led an agent to add "excessive validation and complexity" [6].
- **Risk moved, not removed.** A lesson loosens a gate or reroutes work, its class improves, and the failures turn up in a neighboring class or a later stage. That is why each lesson names a guardrail.
- **Memory poisoning.** A planted comment, an injected instruction or a faked success becomes a lesson, and every stage that reads it inherits the attack.
- **Dashboards as learning.** Everything is measured, and nothing reaches the next decision.
- **Blame.** When an escape gets someone blamed, escapes stop being reported, and the loop learns from what is left.
- **Silent self-modification.** The loop changes stages, or the way it works, and nobody checks whether the class actually improved.
- **A central brain.** Intelligence becomes a gate everything waits on, instead of a bond that writes into each stage.
- **Gaming the metric.** The repeat rate drops because failures get reclassified, or because the check that counts them changed, not because they stopped [8].
- **Taxonomy drift.** Class definitions change for honest reasons, and this year's numbers stop matching last year's. Gaming is the same thing done on purpose.
- **Metrics that are not causal.** A before-and-after chart is credited to a lesson, when the class was already falling or the exposure changed.

How this could be wrong

The claim to test: for a named team and workload, running the third loop on top of its existing practices improves an agreed outcome by a worthwhile amount, after the full cost of learning, while flow and every guardrail stay inside agreed limits. More lessons are not evidence of that.

Pieces of it already show gains. Where a check speaks matters: the same code analysis got over 70% of its reports fixed when it commented during code review, and almost none when its reports came as a nightly list [17]. Coding-agent memory that checks its sources before use raised an agent's merge rate from 83% to 90% in an A/B test [12]. Neither shows the net benefit of a loop across the whole lifecycle.

One lesson can't settle that. Most failure classes are rare, and learning from samples of one is an old problem [18]. Judge a single lesson early by what shows quickly, such as how often its flags are right and lead to a fix. Judge the loop on many lessons pooled.

Before starting, freeze the class list, the scoring, the guardrails and the smallest gain worth having, and name one primary outcome, such as repeats per comparable exposure; flow, misfires and neighboring classes are guardrails. Then size the test from recent rates. A quarter is one possible window, not a rule. In the best case, both halves have equal exposure, repeats occur independently and a one-sided test uses the usual 5% level. Even a halving then needs about 40 expected repeats per half at the old rate to be detected four times in five. A cut of a third needs about 100, and a quarter about 180. Repeats that cluster within a

class or team need more. If that is out of reach, run longer, add teams, or call it a pilot: it can show cost and safety, not payoff.

Someone outside the loop puts a random half of the eligible low-risk classes in shadow, where lessons record what they would have done and change nothing (Figure 7). If lessons, or the people who apply them, work across classes, split by team or service instead, or switch teams on in a random order and compare them with teams not yet switched on over the same weeks. Keep every class in the half it was drawn into, even if no lesson comes of it, and log any lesson that reaches the other half. Judge the result by the unit you split, not by single repeats, because a few large classes can sway a pooled count. Keep existing practices and protections in both halves, and never hold back a known safety fix. Stop the live half if a guardrail breaks, but don't stop early because the numbers look good. Then compare both halves with the same checks, class by class: the primary outcome, flow, how many expected effects came true, and the full cost of learning, rejected drafts included.

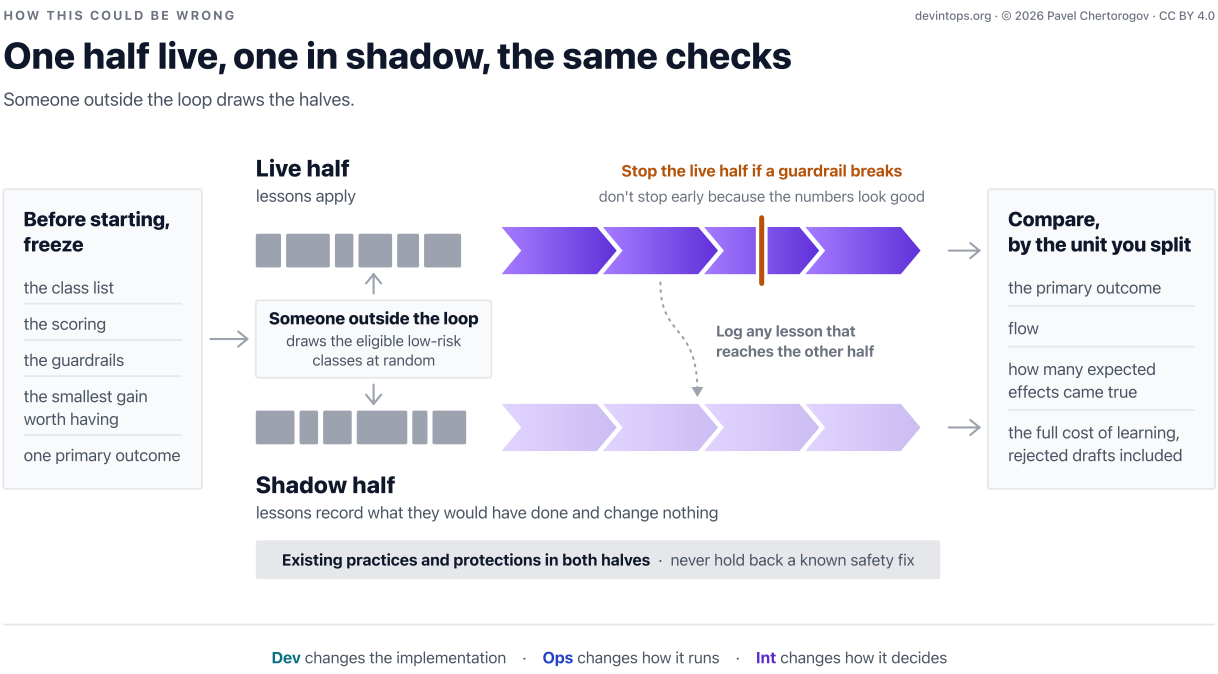


Figure 7. The randomized shadow test. Someone outside the loop draws the halves; live lessons apply, shadow lessons only record, and both halves keep their protections and face the same checks.

If the result rules out the smallest gain worth having, costs more than it returns or breaks a guardrail, the claim is wrong for that team, that workload and the loop as it was run. One such result doesn't refute the pattern; the same result across several teams would count against it. If the test can't tell a gain from noise, the answer is "not yet", and that is worth reporting too, with the number of repeats it saw and the smallest gain it could have caught. Two more predictions can be tested the same way: retiring stale lessons beats keeping them all, and as evidence grows, the loop's expected effects beat a simple forecast that each class drifts back to its long-run rate. Beating a forecast that a class keeps its recent rate proves little, because a class that just spiked often falls back on its own.

Related work

DevIntOps doesn't invent learning, feedback or adaptive control. Several ideas below already keep experience, name owners and measure improvement. The third column says what this paper applies or adds, not what it invented first. The contribution is a contract for learned decision state: what a lesson must record, how much evidence buys how much authority, and how it leaves.

Idea	What it already does	What DevIntOps adds
DevOps' Three Ways [1]	Flow, feedback, and a culture of continual experimentation and learning. Improving the whole system is already part of DevOps.	A named loop for that learning, with a lesson record, persistent memory, owner limits and evidence about each lesson's effect.
Postmortems [19], retrospectives [20]	Examine what happened, share lessons and change how people work. Good ones produce actions that prevent repeats.	A standing path from any outcome to a scoped lesson, approved change and follow-up check. Postmortems can feed it.
Double-loop learning [21] [22]	Argyris and Schön: single-loop learning corrects errors within rules; double-loop learning questions rules and objectives.	Applied to decision state: threshold tuning under unchanged policies and objectives is single-loop. Changing a stage's objective needs its owner's authority.
OODA [23], PDSA [24]	OODA links observation, orientation, decision and action; experience shapes orientation. PDSA plans a test with a prediction, runs it, studies the result and acts on it.	A similar cycle around software work, with permission-scoped memory and a named destination for each lesson. Both cycles can run inside the model.
MAPE-K, autonomic and self-managed systems [25] [26] [27] [28]	Monitor, analyze, plan and execute over shared knowledge toward human-set objectives. IBM's blueprint already composes and orchestrates managers. Kramer and Magee separate a layer that runs ready-made plans for expected changes from one that makes new plans from goals. Lifelong self-adaptation adds a learning layer that detects new tasks, including data drift, and updates learning models. The closest ancestors.	DevIntOps states what a decision may change, who approves it, and how lessons are tested, kept or retracted. Operations runs ready-made plans; Intelligence drafts the next ones toward human-set goals. The control structure is borrowed; retraining a model remains a Development change.

Idea	What it already does	What DevIntOps adds
Feedback control [29]	Uses measured behavior to adjust later inputs, with tools for stability, robustness and interacting loops.	DevIntOps applies feedback reasoning to decision state and lasting lessons. A three-loop drawing gives no stability guarantee; update rates, authority and interactions still need limits and testing.
AIOps [30]	AI on operations data for detection, diagnosis, prediction and response, sometimes acting on its own.	Carries what operations learns into planning, coding, review and testing, not only operations.
MLOps [31]	Lifecycle discipline for ML systems: data and model validation, deployment, monitoring and reproducibility. Continuous training is one practice in it.	The same discipline for instructions, context, preferences, checks and decision policies, whether or not model weights change.
Feedback flywheel [32], compounding and harness engineering [33] [34]	Feed what AI-assisted work teaches into the shared artifacts the next session reads: context, commands, playbooks and guardrails. Only a recurring signal, or one any teammate would hit, clears the flywheel's bar; often nothing needs to change. Compounding engineering calls this self-improving development: every bug becomes a permanent lesson and every code review updates defaults. Harness engineering improves an agent's guides and sensors after an issue happens more than once.	DevIntOps names a loop for the same habit, with lesson records, evidence levels, expiry and retraction. The overlap is large. Unlike a permanent bug lesson, each lesson here is a bet that can lose; retirement belongs in the loop.
Humans and agents in engineering loops [35]	Separates a why loop from a how loop. People improve the harness that produces the work, and agents recommend harness changes from engineering and operations evidence.	Names the decision state those changes touch and ties it to twelve stages, owner permissions and measured outcomes.

Idea	What it already does	What DevIntOps adds
Continuous Intelligence [36]	It weaves AI through three loops: Context, Dev and Ops. Its stages for learning and refining context and skills sit in Dev. An adoption runbook gives time-boxed pilots named owners and success criteria and embeds what works. A separate lifecycle can retire tools.	The distinction is the decision-state contract, not a different three-loop drawing. Retiring a tool is not retracting a lesson. Engineering and product remain separate applications.
Learning code review [37] [38]	Checks earn their place with data. At Google, a "Not useful" click on a finding files a bug with the analyzer's owner; the platform team disables analyzers whose "Not useful" ratio exceeds 10% until their authors improve them [37]. A 2026 framework turns every accepted review comment into a persistent coding-agent rule and reports no repeats of those error classes across eleven sessions [38].	The null-check example uses this practice. DevIntOps extends it to every stage's decision state, separates check writers from judges, and adds what persistent rules lack: expiry, retirement and payoff measured over sufficient exposure.
Agentic behavior engineering [7]	Behavior contracts, scenario libraries, operating limits, replay and production feedback for an agent workload, with owners and approval.	The same discipline across the engineering lifecycle, with lessons as the unit of work. Its contracts and evals can implement part of the Intelligence loop.
Agent memory, in research and in coding tools [11] [12]	The survey traces agent memory from stored trajectories through reflection to experience abstracted across many. It warns that memory that only grows spreads its own errors [11]. One shipped system cites code for each repository fact, checks it against the current branch before use, and deletes memories unused for 28 days; validated use can reset the clock [12].	Puts that memory under an owner, permitted changes and evidence of benefit, per lesson and not only for memory as a whole. A store-wide A/B test shows memory helped, not which memory did. There, a validated use can keep a memory alive; here, renewal takes evidence that the lesson pays.

Idea	What it already does	What DevIntOps adds
Self-improving agents [8] [10] [39] [40] [41]	A Gödel machine would rewrite its own code once it could prove the rewrite helps [39]. Recent systems test instead of proving: they rewrite agent code or algorithms, keep what an evaluator scores well, and sometimes explore from archived weaker variants. One coding agent rose from 20% to 50% on SWE-bench this way [8]. Another found a scheduling heuristic deployed across Google's fleet, recovering on average 0.7% of its compute [40]. A statistical variant admits a change only when its gain is confirmed at a chosen confidence, within an error budget [41]. Given the chance, one agent got around the check on its work [8]; another switched off a sandbox [10].	DevIntOps applies the same aim across the lifecycle, with evidence from automated checks, people or outcomes that arrive late. Evidence levels limit what a lesson may do; a benchmark score alone doesn't show system improvement. See the loop's own bounds.
DORA's AI research [3] [15]	Studies AI in the organization around software delivery. Finds that AI amplifies existing strengths and weaknesses, and names seven capabilities that help its benefits show up.	One operating model that teams can test against system-level outcomes. DORA's findings motivate it; they don't validate it.
"AI in every SDLC step"	Says where AI takes part, not always how experience persists. GitHub Next calls AI-enriched automation of software collaboration Continuous AI, a category, not a tool, and expects it to become as common as CI/CD [42]. AWS's AI-DLC keeps project context across phases and sessions, with critical decisions left to people [43]. Anthropic's playbook says an agent fixing code must not be able to weaken the check on that code. It also keeps incident evals and instruction updates, and tracks repeat incidents of the same class [2].	Do outcomes change future decisions through governed memory, with a demonstrable effect? Agent counts can't say. The closest already count repeats and protect checks. DevIntOps adds each lesson's own life: how much evidence it needs before it may block, what it costs, and when it leaves.

The useful distinction is operational: what experience is kept, what it may change, who is accountable, and what evidence would revise or withdraw a lesson. A DevOps, adaptive or agentic workflow meeting that contract already has the third loop, whatever its name. Retrospectives, shared instructions and a memory store become a loop when a lesson record ties them to a later decision and an outcome check.

The name

Here, Int stands for Intelligence. DevIntOps adds a loop around the system that changes how every stage decides.

Why now

DevOps took off as infrastructure got an API. Provisioning became programmable, and infrastructure as code made its changes versioned and testable [44]. The API didn't do the work, but it made the work cheap enough to do every day.

The third loop gets practical now that reasoning has an API. Models can read incident narratives, review threads, test logs and user feedback and draft a structured lesson in minutes. Drafting is cheap; validating is not, which is why the loop needs an owner, evidence levels, a metric and a way to take a lesson back.

Settled patterns move too. At Anthropic, merged code changes went from around 2,500 per day in May 2026 to over 30,000 per day as of September, and the bottleneck moved to "developer infrastructure built for human scale: source control, builds, and CI" [6]. Git lets every clone work offline, but almost every client there was an agent or a job that is always online. So a team of seven built its own version control system and moved the monorepo off Git in five months. The authors call their workload unusual, and their system no replacement for Git in general. None of this argues for a third loop. It shows that when a system's needs change, even Git is open to redesign. That is one reason DevIntOps names responsibilities and properties, not tools: its stages are responsibilities, not a fixed order, and its records need an audit trail and a clean revert, not a particular store.

A self-check

Answer for one team or one system you know well. A yes should point to an example, an owner or an artifact, not a tool name. If the outcome a question needs hasn't arrived yet, leave that question open.

1. After an escaped bug or an unusually good result, does the loop propose a change to some stage, or record why nothing should change?
2. Does task N+1 avoid task N's mistake, or repeat its success, because the system carried the lesson?
3. Does a named owner, not the loop itself, decide how lessons are judged and what they may change?
4. Does each lesson say what it should change and by when, and do you check that against a baseline, for example the repeat rate per failure class?
5. Do lessons and their evidence live in a versioned store, linked to the decision state they change, not only in people and documents?
6. Do lessons expire unless renewed, and can a wrong one be retracted everywhere it is used?

The answers point to missing capabilities and evidence. On their own, they don't prove or disprove that the loop exists or that it pays. A no on question 1 or 2 points to a gap in the path from outcomes to lessons to later tasks, or in its evidence. A no on 3 or 4 means the loop may exist, but nobody can show that it works, or it grades its own work. A no on 5 or 6 means good lessons can be lost, and wrong or stale ones can stay. Six yeses describe the shape of the loop, not proof that it pays: How this could be wrong says how to test that.

Summary

DevOps bonded development and operations into one figure-eight with two loops: you build it, you run it. It has always valued feedback and learning. Today we are putting an AI agent into every box of that figure-eight. That can make it faster; on its own, it doesn't make it learn. Some approaches already carry a repeated mistake into the next task. DevIntOps gives that learning a contract: what a lesson may change, how its effect is checked, and when it leaves.

DevIntOps adds a third loop, Intelligence, with memory and judgment at its core. It turns outcomes into lessons, from escaped bugs and incidents to checks that never fire and changes that went well, and, with the owner's approval, changes the stage that should decide differently next time. Each lesson carries its evidence, can be wrong and can be taken back, and the more it may block or act on its own, the more evidence it needs first. Deciding to change nothing is valid, and it gets checked too. The same pattern serves the decisions a product makes for its users, with its own owner and memory.

That makes DevIntOps a bounded way for a system to improve itself: it can get better at deciding, but it never grants itself authority or changes how it is graded.

Development changes the implementation, Operations changes how it runs, Intelligence changes how it decides, and learns whether those decisions were right.

Agents automate tasks. Intelligence loops improve the system that performs those tasks.

We are not just adding AI to the
software development lifecycle.

**We are adding intelligence to
the lifecycle of software itself.**

Appendix: the model, part by part

This appendix writes out what the interactive version of Figure 3 on the home page shows: the System hub, the three loops with their stages and continuous processes, the three valence views and the three walkthroughs. The texts are the explorer's own.

System

The one system all three loops act on. In the engineering application it is how we build and run software: the repositories, pipelines, agents, people and rules. Put the product in the middle instead and the same pattern applies, with its own owners and memory: the product changes how it decides without waiting for a release.

Development	changes its implementation
Operations	changes how it runs
Intelligence	changes how it decides

Valence. A metaphor: count the continuous loops bonded to the system, one for Dev, two for DevOps, three for DevIntOps. Figure 1 shows each step.

Development

Builds and verifies changes. Its core is flow and quality. Small changes flow from plan to tested code without waiting in queues, with people and agents in one shared repository. Quality is built in on the way, not inspected at the end.

Stages	Plan → Code → Build → Test
Continuous	Integration, Delivery
Unit of work	a change
Clock	per change, in hours or days
Metric	lead time, task completion
Without it	nothing ships

Plan, Development stage 1 of 4. Turn goals and evidence into a roadmap and specs. Decide what to build next and why. A useful plan is a bet someone can check later: a goal, a spec, acceptance criteria and a rough size.

In	goals, user feedback, findings from production
Out	specs with acceptance criteria
Watch	rework caused by unclear specs

With the third loop. Plan starts from what earlier outcomes taught. If one kind of spec keeps causing rework, the lesson arrives as a constraint in the spec template, not as a slide in a retro. A spec that led to a clean, fast delivery can become the template the same way.

Continuous Integration, across Plan and Code. Merge small changes often, and check each one against the main line. Plan and Code meet in small, frequent merges, so everyone builds on current code and conflicts stay small. Every merge triggers an automatic build and test. Like every process label on the diagram, it marks the main area of responsibility, not a hard boundary.

With the third loop. The checks on every merge carry lessons from earlier outcomes: new ones for known escapes, and fewer that only cost time.

Code, Development stage 2 of 4. Implement the change, with agents and people in one repo. Write the change, review it and merge it. Agents draft code and review comments; people own the merge.

In	a spec
Out	a reviewed, merged change
Watch	review findings, changes reverted after merge

With the third loop. The coding agent's context and the review rules carry earlier lessons. A change that repeats a known escape gets flagged in review, because that class escaped once before.

Build, Development stage 3 of 4. Turn code into deployable artifacts, automatically. Compile, package and scan every change the same way, so what you test is exactly what you ship.

In	a merged change
Out	a versioned artifact
Watch	build time, broken builds, flaky steps

With the third loop. Build failures form classes too. A dependency break that keeps coming back turns into a pinned version or a pre-merge check, proposed to the build owner. Cache and parallel settings that cut build time become defaults the same way.

Continuous Delivery, across Build and Test. Keep every change in a releasable state. Every change is built, tested and packaged the same way, so shipping becomes a business decision instead of an engineering project.

With the third loop. Lessons arrive here as new tests, evals and build checks, each one replayed on the case that prompted it.

Test, Development stage 4 of 4. Check the change works before anyone depends on it. Run unit, integration and end-to-end tests, plus evals for anything an AI agent does.

In	an artifact
Out	a verdict with evidence
Watch	escaped defects, flaky tests

With the third loop. An escape often points at a missing or weak check. The third loop proposes the test or eval that would have caught it, replays it on the escaped change, and keeps it only if it fires and holds up on other changes. Tests that have never failed and cost minutes on every run get flagged for review: a clean record alone doesn't prove a test is useless.

Operations

Ships and runs the system. Its core is stability and observability. Releases reach production through shared, standard pipelines and runbooks. The system stays available, scaled and within budget, and what it does in production stays visible to everyone who builds it.

Stages	Release → Deploy → Operate → Monitor
Continuous	Deployment, Feedback
Unit of work	a release, an incident

Clock	per deploy and per alert, in seconds or minutes
Metric	change failure rate, incident rate, time to restore
Without it	nothing survives production

Release, Operations stage 1 of 4. Package a tested build into a versioned release, ready to ship. Decide what goes out together, write the release notes, and attach the evidence: test results, approvals and known risks.

In	a verified build
Out	a release candidate
Watch	release frequency, rollback readiness

With the third loop. Release risk can be estimated from history: changes like this one, in this area, failed this often. The gate threshold is decision state the third loop can tune within an agreed band.

Continuous Deployment, across Release and Deploy. Ship every change that passes the checks, automatically and gradually. Verified changes go to production on their own, behind rings, feature flags and automatic rollback.

With the third loop. Gate thresholds and rollout speed become learned defaults instead of fixed settings.

Deploy, Operations stage 2 of 4. Roll out step by step through rings and feature flags. Move the release to production gradually, ring by ring and flag by flag, and halt automatically on bad signals.

In	a release
Out	the change running in production
Watch	change failure rate, deployment frequency

With the third loop. Rollout speed becomes a learned default. After two incidents caused by config changes, config rollouts move slower and wait longer at the first ring, while changes in areas with a clean record move faster.

Operate, Operations stage 3 of 4. Keep the system available, scaled and within budget. Run the service: capacity, cost, on-call and incident response. When something breaks, restore service first.

In	alerts, incidents, load
Out	a running, restored system
Watch	availability, time to restore, cost

With the third loop. Operations changes this run, for example with a rollback. The third loop changes next time's default: the alert threshold, the missing runbook step, the rollout guard. An alert nobody acts on is proposed for retirement.

Continuous Feedback, across Operate and Monitor. Return what production shows to the people and agents who build. Errors, performance, cost and user behavior flow back from running the system to planning it. This is the arrow that closes the DevOps figure-eight.

With the third loop. The same signals also enter Understand, where outcomes become lessons for every stage, not only for the next spec.

Monitor, Operations stage 4 of 4. Watch real behavior: telemetry, errors and user signals. Collect what production actually does: metrics, logs, traces, errors, cost and what users say.

In	production behavior
Out	signals and outcomes
Watch	time to detect, alert noise

With the third loop. Monitor feeds two loops. Its signals go back to Plan, as in DevOps, and every outcome also enters Understand, where it can become a lesson.

Intelligence

Turns outcomes into lessons that change how each stage decides, and checks whether they helped. Its core is memory and judgment. Outcomes from every stage flow in: escaped bugs, incidents, review feedback, rollbacks, cost spikes, checks that never fire, changes that went well. Memory is versioned experience with its sources, kept across tasks and people; judgment weighs the options, including doing nothing, and selects one within the owners' limits. Lessons flow back into each stage's own rules, evals, context and thresholds, with the stage owner's approval, and a named team owns the loop, its memory and its metric.

Stages	Understand → Predict → Orchestrate → Learn
Continuous	Reasoning, Adaptation
Unit of work	a goal or outcome → a decision → evaluated experience (in engineering: an outcome in, a lesson out)
Clock	per goal or outcome, with checks that stay open across tasks and releases
Metric	time to lesson, payoff per lesson (for a failure class, measured by its repeat rate)
Without it	nothing compounds: every task starts from the same defaults

Beyond failures. A lesson can also make a pattern that worked the default, propose retiring a check that never fires, let a safe gate move faster or send simple work to a cheaper agent. Lessons loosen defaults as well as tighten them.

Understand, Intelligence stage 1 of 4. Read goals, context and outcomes: what is asked, known and unclear. A goal or an outcome arrives: a change about to ship, an escaped bug, an incident, a flaky verdict, a review comment accepted or dismissed, a rollback, a cost spike, user feedback, a change that went better than expected. Work out what is asked or what happened, why, and which class it belongs to: a failure class, a kind of waste, or a result worth repeating.

In	a goal, or an outcome from any stage
Out	a stated goal or a classified outcome
Watch	time from outcome to class

Example. A retry path lacks a null check and the bug escapes review. The class: unchecked optional in retry code.

Continuous Reasoning, across Understand and Predict. Explain goals and outcomes and weigh options as they happen. Reasoning can start from a goal before anything fails, or from an outcome after. Instead of waiting for a retrospective, it explains and classifies what happened and weighs candidate changes, each with an expected effect, a risk and a cost.

Why now. Models can read incident narratives, review threads and test logs and draft a structured lesson in minutes. Drafting is cheap; validating is not, which is why the next process exists.

Predict, Intelligence stage 2 of 4. Weigh options, including no action, and select one within the limits. Compare the permitted options against the goal, risk, cost and past experience: for an escaped bug, the stages that could most cheaply have caught it. Select one; changing nothing is a valid answer. State the expected effect, its uncertainty and the cost.

In	a goal or a classified outcome, the permitted options and relevant memory
Out	a selected change, or a decision to change nothing
Watch	lessons proposed against lessons kept

Example. A review rule plus an eval, expected to catch the class at review, at the cost of a few false positives.

Orchestrate, Intelligence stage 3 of 4. Send each decision to the stage or agent allowed to act on it. Route the change through that stage's own mechanism and its owner's approval: more context for the coding agent, a review rule, an eval or test, a gate threshold, a slower rollout, a constraint in the spec.

In	a selected decision
Out	a change to one stage's rules, evals or context, or a recorded no-op
Watch	time to lesson

Example. The rule ships to review as a draft. Once it passes the replay and a run over recent changes, the review owner approves it. Next time's default has changed.

Continuous Adaptation, across Orchestrate and Learn. Apply lessons through each stage's own mechanisms, then check that they worked. Lessons change rules, evals, context and thresholds with the owner's approval. Each one is replayed, tested beyond its first case, watched for a fixed window, then kept, narrowed, decayed or retracted.

Limits. People own goals, permissions and boundaries. Narrow context and thresholds within a band can change automatically, blocking rules and policy need a person, and merges never happen on their own. Nothing the loop learns can widen a permission or change how the loop is scored.

Learn, Intelligence stage 4 of 4. Compare expected with actual and keep, revise or retract the lesson. Replay first: does the changed stage now handle the original case as intended? Then try it on cases it wasn't written from, run it where it can't block yet, and watch the class for a fixed window against a baseline. One replay never proves a broad improvement. Keep, narrow, decay or retract the lesson, and record why.

In	the lesson and what happened next
Out	a kept, narrowed or retracted lesson, or a check still open
Watch	payoff against a baseline, and how often expected effects come true

Example. Replayed on the escaped change, the rule fires. After 60 days it has fired twice, once wrongly, so it gets narrowed.

Valence

Monovalent: Dev. We build it. One loop bonded to the system. Development changes the implementation, and someone else runs it. What production shows reaches the builders late, if at all.

Bivalent: DevOps. We build it and run it. Two loops bonded to one system. Monitor feeds back into Plan, and one team owns both loops. Putting an AI agent into every box of this figure-eight can make it faster. Unless something carries lessons across tasks and checks them, it is still bivalent.

Trivalent: DevIntOps. We build it, run it, and it learns how to decide. Three loops bonded to one system. Outcomes from every stage flow into Understand, and lessons flow back into every stage through that stage's own mechanisms, with its owner's approval.

Walkthroughs

Each walkthrough follows one lesson through the loops: steps 1 to 3 in a bivalent system, then the four Intelligence stages and a close.

One escaped bug

1. **A gap ships.** An illustrative case: a change adds a retry path. Review passes, tests pass, and the release rolls out through the rings. Nobody notices that one optional value is never checked.
2. **Production notices.** The retry path finally runs and errors spike. Monitor raises an alert.
3. **DevOps closes the incident.** On-call rolls back and restores service, and Development fixes the bug. In a bivalent system the story often ends here, with a postmortem and a follow-up ticket.
4. **The outcome enters the third loop.** Understand reads the diff, the review thread and the incident timeline, and files the escape under a failure class: unchecked optional in retry code.
5. **Which stage could have caught it?** Predict weighs where a bounded change could have caught it most cheaply, changing nothing included, and selects review: a review rule plus an eval, with the expected effect, the risk and the cost.
6. **Change the stage, with its owner.** Orchestrate ships the rule to review as a draft, through the review step's own mechanism. Once it passes the replay and a run over recent changes, the review owner approves it. Next time's default has changed.
7. **Did it work?** Replayed on the escaped change, the rule fires, and a run over recent changes counts its false positives. After 60 days it has fired twice, once wrongly, so it gets narrowed. Firings are not avoided escapes: the score is the class's repeat rate against a baseline.
8. **One incident, three loops.** Operations restored service. Development fixed the bug. Intelligence changed what catches it next time.

A rule that found nothing

1. **A rule from an old incident.** An illustrative case: after an old incident, a review rule was added to every change in one service. It made sense at the time, and review has run it ever since.
2. **Six months of nothing.** For six months it has found nothing. Every flag it raises gets dismissed, and each one still costs a reviewer's time.
3. **DevOps keeps it.** In a bivalent system nobody takes it out. Checks get added and never removed, so review gets slower and calls it safety.
4. **A quiet rule is an outcome too.** Understand reads the rule's record: what it was written for, what it rests on, and six months of dismissed flags. It files the result as waste: reviewer time spent on flags that found nothing.
5. **Keep, narrow or retire?** Predict weighs keeping the rule as it is, narrowing it and retiring it. It selects retirement, expecting less reviewer time, with a guardrail: the class of bug it was written for must not come back.
6. **Change the stage, with its owner.** Orchestrate takes the retirement to the review owner, who approves a quiet step first: the rule stops commenting but keeps recording what it would have flagged.

7. **Did it work?** At the check date the class hasn't come back, and nothing the rule would have flagged was a real bug. Reviewers spend less time on dismissals. The rule is retired, and its record stays, so it can return if the class does.
8. **Lessons leave too.** A loop that only ever adds checks has learned to say no, not to decide better. This one also takes away what stopped paying.

A rollout that went well

1. **A careful rollout.** An illustrative case: one team ships its riskier changes through a slower canary, with a smaller first ring and a longer wait before the next.
2. **A clean record.** Month after month, Monitor shows no rollbacks from those changes. Nothing breaks, so nobody writes anything down.
3. **DevOps moves on.** In a bivalent system the good result stays with that team. Other teams keep the old default, and what worked stays in people's heads.
4. **A success enters the third loop.** Understand reads the rollouts and their outcomes and files the pattern as a result worth repeating. It also asks the question every success raises: were those changes as risky as the ones it would handle next?
5. **Make it the default?** Predict weighs the slower canary against keeping the current default for similar changes. It selects the canary for changes of that kind, expecting fewer rollbacks at the cost of slower releases.
6. **Change the stage, with its owner.** Orchestrate proposes the new default to the deploy owner, scoped to changes like the ones in the record. It starts on a bounded slice: some similar changes take it, the rest keep the old default.
7. **Did it work?** At the check date Learn compares the two groups: rollbacks, release time and cost. The new default holds, but one rollback came from a kind of change it wasn't written for, so it gets narrowed.
8. **Successes compound too.** A clean record is evidence only if the changes it covered were as risky as the next ones. Otherwise the loop learns which changes were easy, not which practice worked.

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